

Homework 2

EE 999 — Introduction to Control Systems

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Due Wednesday, September 23, 2026

Problem 1.A.2.1 — What is a "free-body diagram"?

A free body diagram is a diagram that shows the forces acting on an object or set of objects.

Problem 1.A.2.2 — What are the two forms of Newton's law?

Translational motion: $F = ma$; Rotational motion: $M = I\alpha$

Problem 1.A.2.3 — For a structural process to be controlled, such as a robot arm, what is the meaning of "collocated control" and "Noncollocated control"?

Collocated control means the sensor and actuator are rigidly attached to one another, whereas there is flexibility between the actuator and sensor in a noncollocated system.

Problem 1.A.2.4 — State Kirchhoff's current law.

Kirchhoff's current law states that current entering any given electrical node must equal current exiting a node.

Problem 1.A.2.5 — State Kirchhoff's voltage law.

Kirchhoff's voltage law states that the sum of all voltages around a closed circuit must equal zero.

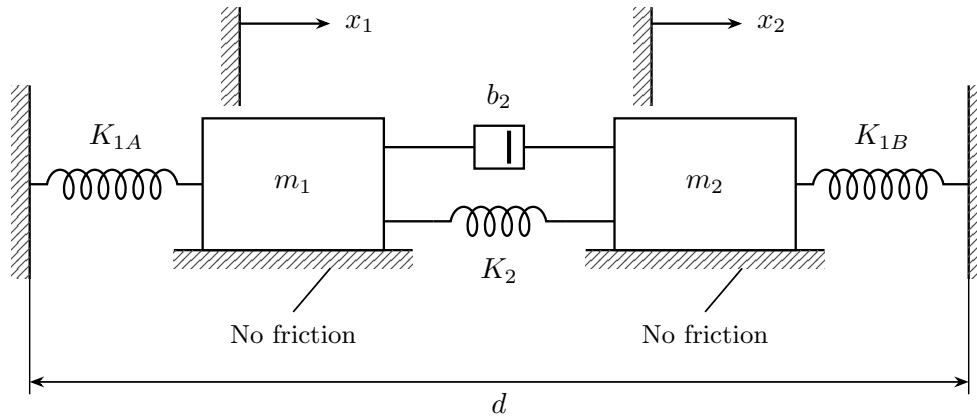
Problem 1.B — Differential equation for mechanical system

Figure 1: The two-mass system of Problem 1.B.

Signal	Units	Description
$x_1(t)$	[m]	Position of mass m_1
$x_2(t)$	[m]	Position of mass m_2

Table 1: Signals of Problem 1.B.

Parameter	Units	Description
m_1	[Kg]	mass of m_1
m_2	[Kg]	mass of m_2
K_{1A}	[N/m]	Outer spring stiffness, wall to m_1
K_{1B}	[N/m]	Outer spring stiffness, m_2 to wall
K_2	[N/m]	Inner spring stiffness
b_2	[N-s/m]	Shock absorber

Table 2: Parameters of Problem 1.B.

Constituent equations:

$$\begin{aligned}
 f_{K_{1A}}(t) &= K_{1A}x_1(t) \\
 f_{K_{1B}}(t) &= K_{1B}(-x_2(t)) \\
 f_{K_2}(t) &= K_2(x_2(t) - x_1(t)) \\
 f_{b_2} &= b_2(\dot{x}_2(t) - \dot{x}_1(t)) \\
 f_{net_{m1}}(t) &= m_1\ddot{x}_1(t) \\
 f_{net_{m2}}(t) &= m_2\ddot{x}_2(t)
 \end{aligned}$$

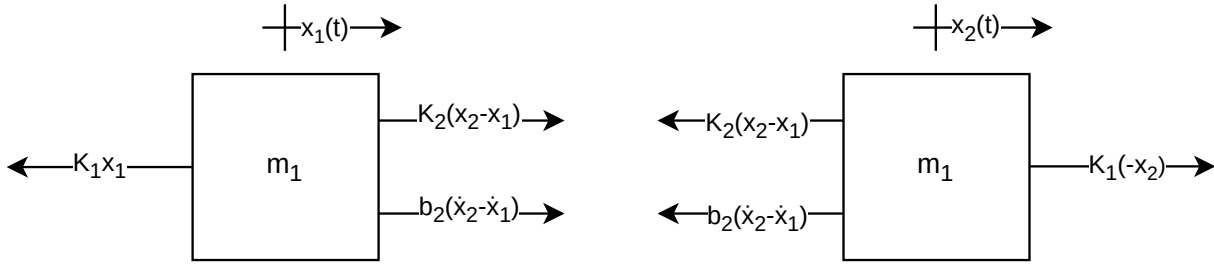


Figure 2: Free body diagram for problem 1.B

Continuity Equations:

$$K_1 = K_{1A} = K_{1B}$$

$$m = m_1 = m_2$$

$$\begin{aligned} m\ddot{x}_1(t) &= f_{K_2} + f_{b_2} - f_{K_{1A}} \\ &= K_2(x_2 - x_1) + b_2(\dot{x}_2 - \dot{x}_1) - K_1x_1 \\ &= K_2x_2 - K_2x_1 + b_2\dot{x}_2 - b_2\dot{x}_1 - K_1x_1 \\ &= -b_2\dot{x}_1 - (K_1 + K_2)x_1 + b_2\dot{x}_2 + K_2x_2 \end{aligned}$$

$$\begin{aligned} m\ddot{x}_2(t) &= f_{K_{1B}} - f_{K_2} - f_{b_2} \\ &= K_1(-x_2) - K_2(x_2 - x_1) - b_2(\dot{x}_2 - \dot{x}_1) \\ &= -K_1x_2 - K_2x_2 + K_2x_1 - b_2\dot{x}_2 + b_2\dot{x}_1 \\ &= -b_2\dot{x}_2 - (K_1 + K_2)x_2 + b_2\dot{x}_1 + K_2x_1 \end{aligned}$$

Standard form:

$$m\ddot{x}_1(t) + b_2\dot{x}_1(t) + (K_1 + K_2)x_1(t) = b_2\dot{x}_2(t) + K_2x_2(t) + f(t)$$

$$m\ddot{x}_2(t) + b_2\dot{x}_2(t) + (K_1 + K_2)x_2(t) = b_2\dot{x}_1(t) + K_2x_1(t)$$

*NOTE: $f(t)$ has been added to the m_1 equation so the transfer function could be created

Frequency domain:

$$(ms^2 + b_2s + (K_1 + K_2))X_1(s) = (b_2s + K_2)X_2(s) + F(s)$$

$$(ms^2 + b_2s + (K_1 + K_2))X_2(s) = (b_2s + K_2)X_1(s)$$

Find $X_1(s)$ in terms of $X_2(s)$ so substitution may be performed:

$$X_1(s) = \frac{ms^2 + b_2s + (K_1 + K_2)}{b_2s + K_2}X_2(s)$$

Then perform the substitution:

$$\begin{aligned}(ms^2 + b_2s + (K_1 + K_2)) \frac{ms^2 + b_2s + K_1 + K_2}{b_2s + K_2} X_2(s) &= (b_2s + K_2)X_2(s) + F(s) \\ (ms^2 + b_2s + (K_1 + K_2))^2 X_2(s) &= (b_2s + K_2)^2 X_2(s) + (b_2s + K_2)F(s) \\ ((ms^2 + b_2s + (K_1 + K_2))^2 - (b_2s + K_2)^2) X_2(s) &= (b_2s + K_2)F(s)\end{aligned}$$

Next, gather the terms:

s^4	s^3	s^2	s^1	s^0
m^2	mb_2	$m(K_1 + K_2)$		
	mb_2	b_2^2	$b_2(K_1 + K_2)$	
		$m(K_1 + K_2)$	$b_2(K_1 + K_2)$	$K_1^2 + 2K_1K_2 + K_2^2$
		$-b_2^2$	$-b_2K_2$	
			$-b_2K_2$	$-K_2^2$

Then, combine them to write the entire transfer:

$$G_{FX_2(s)} = \frac{X_2(s)}{F(s)} = \frac{b_2s + K_2}{m^2s^4 + 2mb_2s^3 + 2m(K_1 + K_2)s^2 + (2b_2(K_1 + K_2) - 2b_2K_2)s + K_1^2 + 2K_1K_2}$$

Check Units:

$$\begin{aligned}& \frac{\frac{Ns}{m}s^{-1} + \frac{N}{m}}{Kg^2s^{-4} + Kg\frac{Ns}{m}s^{-3} + Kg\frac{N}{m}s^{-2} + \frac{Ns}{m}\frac{N}{m}s^{-1} + (\frac{N}{m})^2} \\& \frac{\frac{N}{m}s^{-1} + \frac{N}{m}}{\frac{N^2}{m^2}s^{-4} + \frac{N}{m}\frac{N}{m}s^{-3} + \frac{N}{m}\frac{N}{m}s^{-2} + \frac{N}{m}\frac{N}{m}s^{-1} + (\frac{N}{m})^2} \\& \frac{\frac{N}{m}}{\frac{N^2}{m^2}} = \frac{m}{N}\end{aligned}$$

This makes sense because the input is a force and the output is a displacement, so the units of the transfer function should be distance over force.

The difference in velocities between m_1 and m_2 will reach a steady state because b_2 scales with velocity. Since there is no friction, the output $x_2(t)$ will not decay.

Problem 1.C — Modified cruise control

Equation 2.4 from Example 2.1 in the text book reads:

$$\dot{v} + \frac{b}{m}v = \frac{u}{m}$$

Substituting $u = K(v_r - v)$, the equation becomes:

$$\dot{v} + \frac{b}{m}v = \frac{K(v_r - v)}{m}$$

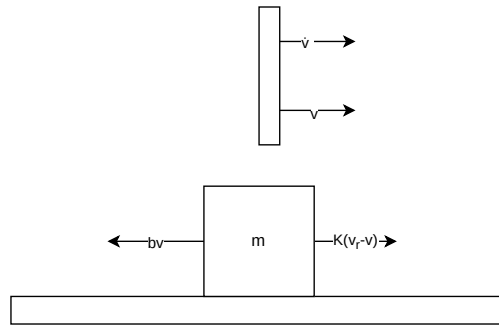


Figure 3: Schematic diagram of car cruise control model

Signal	Units	Description
$v(t)$	[m]	Measured velocity of car
$v_r(t)$	[m]	Commanded velocity of car

Table 3: Signals of Problem 1.B.

Parameter	Units	Description
m	[Kg]	mass of car
b	[N-s/m]	Outer spring stiffness, wall to m
K	[N-s/m]	gain constant

Table 4: Parameters of Problem 1.B.

constituent equations:

$$\begin{aligned} f_b &= bv \\ f_K &= K(v_r - v) \\ f_{net} &= m\dot{v} \end{aligned}$$

Continuity equations:

$$\begin{aligned} f_{net} &= f_K - f_b \\ m\dot{v}(t) &= K(v_r(t) - v(t)) - bv(t) \end{aligned}$$

Standard form:

$$Kv_r(t) = m\dot{v}(t) + Kv(t) + bv(t)$$

$$Kv_r(t) = m\dot{v}(t) + (K + b)v(t)$$

Frequency domain:

$$KV_r(s) = (ms + (K + b))V(s)$$

Transfer function:

$$G_{V(s)} = \frac{V(s)}{V_r(s)} = \frac{K}{ms + (K + b)}$$

Units:

$$\frac{\frac{N \cdot s}{m}}{Kg \cdot s^{-1} + \frac{N \cdot s}{m} + \frac{N \cdot s}{m}}$$

$$\frac{\frac{N \cdot s}{m}}{\frac{N \cdot s}{m} \cdot s^{-1} + \frac{N \cdot s}{m} + \frac{N \cdot s}{m}}$$

Code: hw_02_1C.m

```

1  if exist('OCTAVE_VERSION', 'builtin')
2      pkg load control;
3  end
4
5  m = 1500;          % mass of the car
6  b = 70;           % friction coefficient
7
8  K = 1000;
9  sys1000 = tf(K, [m, K+b]);
10
11 K = 5000;
12 sys5000 = tf(K, [m, K+b]);
13
14 K = 10000;
15 sys10000 = tf(K, [m, K+b]);
16
17 figure;
18 hold on;
19 step(sys1000, sys5000, sys10000)
20 xlabel('time [s]');
21 ylabel('v(t) [m/s]');
22 title('Unit steps for various values of K');
23 legend('K=1000', 'K=5000', 'K=10000', 'location', 'southeast');
24 ylim([0 1.1]);
25 hold off;

```

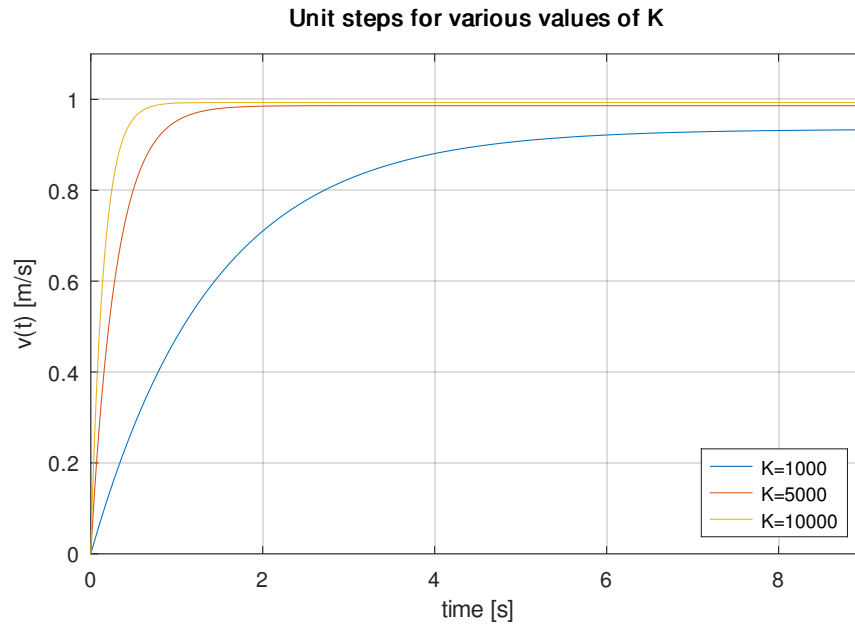
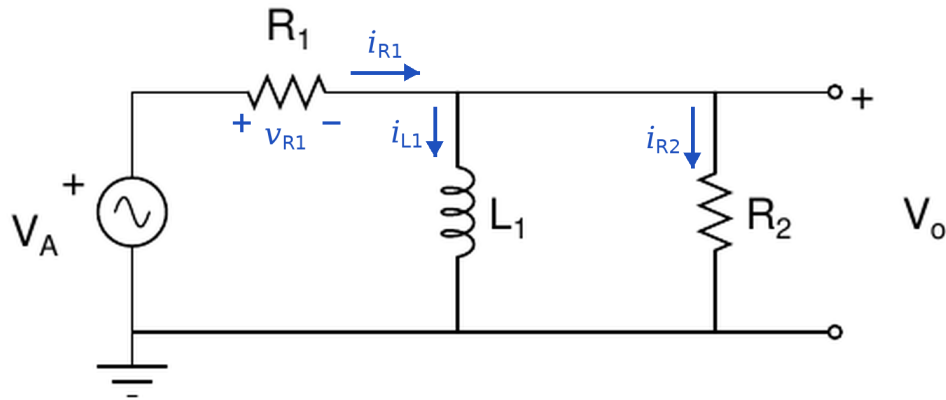


Figure 4: Unit step response of $G_{V(s)}$ for $K = 1000$, $K = 5000$, and $K = 10000$.

No value of K will cause issues with stability or oscillations, but the tradeoff is excessive acceleration against accuracy of output. Given the graphs, a value between 1000 and 5000 would probably get close enough to the target without accelerating too quickly.

Problem 2 — Transfer function of a circuitFigure 5: The RL circuit of Problem 2, with input $v_A(t)$ and output $v_o(t)$.

Signal	Units	Description
$v_A(t)$	[V]	Input Voltage
$v_o(t)$	[V]	Position of mass m_2
$v_{R1}(t)$	[V]	Voltage Across R1
$i_{R1}(t)$	[A]	Current through R1
$i_{R2}(t)$	[A]	Current through R2
$i_{L1}(t)$	[A]	Current through L1

Table 5: Signals of Problem 1.B.

Parameter	Units	Description
R_1	[Ohm(V/A)]	Resistor
R_2	[Ohm(V/A)]	Resistor
L_1	[H(V-s/A)]	Resistor

Table 6: Parameters of Problem 1.B.

Constituent Equations:

$$v_A(t) = v_{R1}(t) + v_o(t)$$

$$i_{R1}(t) = i_{L1}(t) + i_{R2}(t)$$

$$v_o(t) = L_1 \dot{i}_{L1}(t)$$

Continuity Equations:

$$\begin{aligned}
 i_{R_1}(t) &= i_{L_1}(t) + i_{R_2}(t) \\
 i_{R_2}(t) &= \frac{v_o(t)}{R_2} \\
 i_{R_1}(t) &= \frac{v_A(t) - v_o(t)}{R_1} \\
 \frac{v_A(t) - v_o(t)}{R_1} &= i_{L_1}(t) + \frac{v_o(t)}{R_2} \\
 v_o(t) &= L_1 \dot{i}_{L_1}(t)
 \end{aligned}$$

Move things around to get to standard form:

$$\begin{aligned}
 R_2(v_A(t) - v_o(t)) &= R_1 R_2 i_{L_1}(t) + R_1 v_o(t) \\
 R_2 v_A(t) &= R_1 v_o(t) + R_2 v_o(t) + R_1 R_2 i_{L_1}(t) \\
 v_o(t) &= L_1 \dot{i}_{L_1}(t)
 \end{aligned}$$

Frequency Domain:

$$\begin{aligned}
 R_2 V_A(s) &= (R_1 + R_2) V_o(s) + R_1 R_2 I_{L_1}(s) \\
 V_o(s) &= L_1 s I_{L_1}(s)
 \end{aligned}$$

Perform necessary substitutions:

$$\begin{aligned}
 V_o(s) &= L_1 s I_{L_1}(s) \\
 I_{L_1}(s) &= \frac{1}{L_1 s} V_o(s) \\
 R_2 V_A(s) &= (R_1 + R_2) V_o(s) + \frac{R_1 R_2}{L_1 s} V_o(s) \\
 R_2 L_1 s V_A(s) &= (L_1 s (R_1 + R_2) + R_1 R_2) V_o(s) \\
 G_{Ao}(s) &= \frac{V_o(s)}{V_A(s)} = \frac{R_2 L_1 s}{(R_1 + R_2) L_1 s + R_1 R_2}
 \end{aligned}$$

Check Units:

$$\frac{\frac{V}{A} \frac{V}{A} \cancel{s}}{\frac{V}{A} \frac{V}{A} \cancel{s} + \frac{V}{A} \frac{V}{A}}$$

The Gain is a voltage over a voltage, so all units should cancel, as they do with this solution.

Problem 3 — Matlab

Code: hw_02_3A.m

```

1  if exist('OCTAVE_VERSION', 'builtin')
2      pkg load control;
3  end
4
5  m = 1000;
6
7  b1 = 5;
8  A1 = -b1/m;
9
10 B = 1/m;
11 C = 1;
12 D = 0;
13
14 cruise_1 = ss(A1,B,C,D);
15
16 b2 = 50;
17 A2 = -b2/m;
18 cruise_2 = ss(A2,B,C,D);
19
20 b3 = 500;
21 A3 = -b3/m;
22 cruise_3 = ss(A3,B,C,D);
23
24 figure;
25 hold on;
26 step(cruise_1, cruise_2, cruise_3)
27 xlabel('time [s]');
28 ylabel('v(t) [m/s]');
29 title('Unit steps for various values of b');
30 legend('b=5', 'b=50', 'b=500', 'location', 'southeast');
31 hold off;

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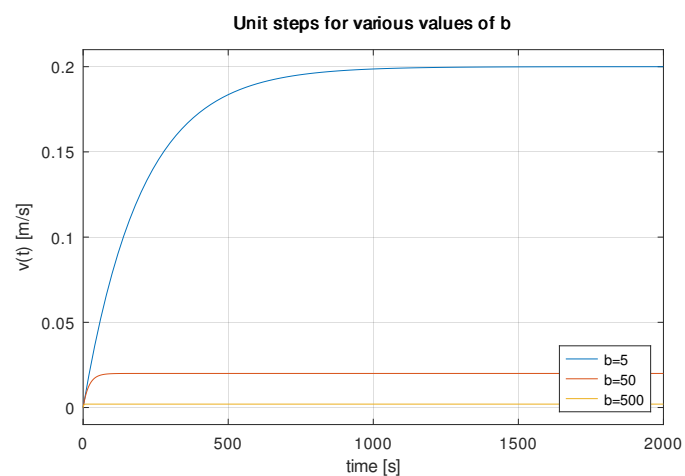


Figure 6: Unit step response of $G_{V(s)}$ for $b = 5$, $b = 50$, and $b = 500$.

Code: hw_02_3A.m

```

1  if exist('OCTAVE_VERSION', 'builtin')
2      pkg load control;
3  end
4
5  m = 1000;
6
7  b1 = 5;
8  A1 = -b1/m;
9
10 B = 1/m;
11 C = 1;
12 D = 0;
13
14 cruise_1 = ss(A1,B,C,D);
15
16 b2 = 50;
17 A2 = -b2/m;
18 cruise_2 = ss(A2,B,C,D);
19
20 b3 = 500;
21 A3 = -b3/m;
22 cruise_3 = ss(A3,B,C,D);
23
24 figure;
25 hold on;
26 step(cruise_1, cruise_2, cruise_3)
27 xlabel('time [s]');
28 ylabel('v(t) [m/s]');
29 title('Unit steps for various values of b');
30 legend('b=5', 'b=50', 'b=500', 'location', 'southeast');
31 hold off;

```

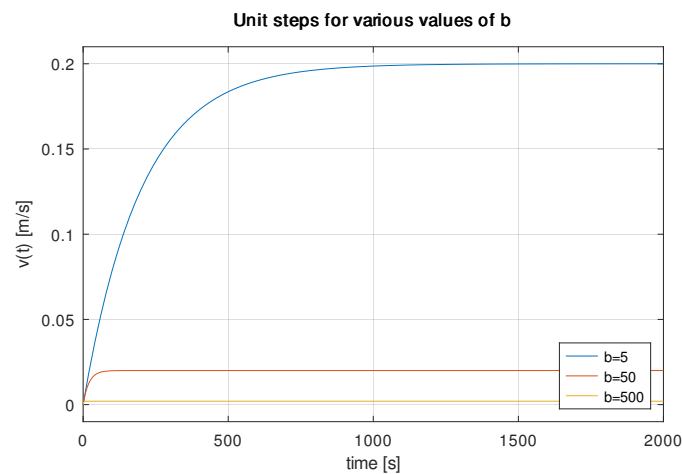


Figure 7: Unit step response of $G_{V(s)}$ for $b = 5$, $b = 50$, and $b = 500$.